

Literature Status: Uniform Spectral Gap for Negative-Parameter Harmonic Measures

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Status

Literature already answered. This note records a later-literature answer to a question suggested by Milman [1]; it is not a new proof from this run.

Source Question

For $n \geq 2$, $|x| < 1$, and $\alpha \geq -n$, let $\mu_x^{n,\alpha}$ be the probability measure on S^n whose density relative to spherical measure is proportional to $|y - x|^{-(n+\alpha)}$. In Section 4.2, page 9 of the source PDF, Milman writes that interpolation between $\alpha = -n$ and $\alpha = 0$ suggests that the spectral gap remains of order n , uniformly for $\alpha \in [-n, 0]$ and $|x| < 1$.

Supporting Answer and Parameter Match

Ma and Wang [2] study $\mu_x^{m,\beta}$ on S^{m-1} , with density proportional to $|y - x|^{-(m+\beta)}$. Their Theorem 1.1 gives, in the branches relevant here,

$$\lambda_1(\mu_x^{m,\beta}) \geq \begin{cases} m-2, & \beta \leq 2-m, \\ \frac{m-\beta}{2} - 1, & 2-m < \beta < -1, \\ \frac{(\beta+3)(m-1)}{4}, & \beta = -1. \end{cases}$$

Set

$$m = n+1, \quad \beta = \alpha - 1.$$

Then $m + \beta = n + \alpha$, so the two papers define the same measure. The displayed estimate becomes

$$\lambda_1(\mu_x^{n,\alpha}) \geq \begin{cases} n-1, & -n \leq \alpha \leq 2-n, \\ \frac{n-\alpha}{2}, & 2-n < \alpha < 0, \\ \frac{n}{2}, & \alpha = 0. \end{cases}$$

Hence

$$\inf_{|x|<1} \lambda_1(\mu_x^{n,\alpha}) \geq \frac{n}{2} \quad (-n \leq \alpha \leq 0),$$

which is exactly the uniform order- n conclusion proposed in the source.

Provenance and Scope

The supporting authors explicitly identify Milman’s work: their abstract says that their general- β bounds improve Milman’s 2015 estimates, and the publisher text discusses Milman’s generalized harmonic measures. Thus this is an explicit later answer, not merely an agent-identified consequence.

The range $\alpha \in [-n, 0]$ is completely covered. Milman’s separate question about $\alpha < -n$ remains outside this identification.

The supporting article has no arXiv version located in the bounded title/author/DOI search, and its publisher PDF endpoint denied access. The source arXiv PDF is included locally; the supporting paper is identified by its stable DOI and exact theorem label.

References

- [1] E. Milman, *Harmonic Measures on the Sphere via Curvature-Dimension*, arXiv:1505.04335; Ann. Fac. Sci. Toulouse Math. 26 (2017), 437–449.
- [2] Y. Ma and X. Wang, *A note on the spectral gap for general harmonic measures on spheres*, Statistics & Probability Letters 141 (2018), 56–61, [doi:10.1016/j.spl.2018.05.022](https://doi.org/10.1016/j.spl.2018.05.022).